Proof Gate and Fair, Sustainable Economics — Evidence Brief

Prepared: 2026-10-30
Scope: Task 81; read-only source research and a bounded commercial-design proposal.
Decision: BLOCKED for genuine-proof validation and report-template reuse.
Economic status: No reward rate, earnings claim, funding source, or backing policy is approved by this brief.

Executive result

The connected-source review found research reports, mining concepts, an older verification test script, and proof-gate UI material. It did not find an owner-approved verifier/circuit/key package, exact compute and storage statements, byte-level public-input encoding, or independent genuine positive proof vectors.

Therefore:

- Keep the genuine-proof control skipped.
- Do not reuse a report template to authorize rewards.
- Treat synthetic adapters, example proof bytes, signatures, attestations, UI simulations, and AI responses as non-evidence for cryptographic verification.
- Do not claim a user can earn a specific amount or that any asset is backed by the reviewed materials.
- Keep existing economic sentinels unchanged. The proposals below are not authorization to alter them.

Evidence matrix

Source • What was observed • Evidence status • Local policy and candidate design • The policy requires exact approved artifacts, bound inputs, genuine compute and storage positives, mutation failures, alternate-encoding replay rejection, and unchanged economic sentinels. Circom + snarkjs/Groth16 is only a design direction. • OBSERVED requirement; no implementation approval. • Google Drive: Crypto Infrastructure research report — https://drive.google.com/file/d/1U00s8xQrFAUBI1dPI0FX4FxIR9-3TntU/view • Discusses proof-of-data concepts, Circom/snarkjs, DePIN, storage, compute, and monetization ideas. It supplies no project circuit, verifier, key, or complete positive vectors. • Research lead, not proof or approval. • Google Drive: verify_lt_stack.py — https://drive.google.com/file/d/1nf7ZZJRVnkDhrlukIveHuyjMRIzWz0F9/view • The inspected proof-related matches include test names and calls for a ZKPassport/Groth16-related feature. They do not establish that the project's compute/storage claims are verified by the required pinned circuit and key. • Test/code reference; not the required positive vectors or approval. • Google Drive: G06 Privacy Approval Review — https://docs.google.com/document/d/1HlWx5pI-uhkvdZvXDoVq6U7wZ05j442md9u8uKkxC4Q/e

dit • Review/checklist material; it does not identify approved compute/storage proof artifacts. • Review material, not proof evidence. • Google Drive: verification.tsx search result • Drive search surfaced this filename, but its contents were not independently inspected in this review. • Unverified lead; no claim is made about its contents. • User-attached React proof-lab inventory • Shows synthetic placeholder proof bytes and a UI function that displays hard-coded PASS/SKIPPED logs rather than invoking the actual policy test command. • UI demonstration only; not a test run or cryptographic evidence. • Google Drive: LTR Research Library • Workbook is discoverable in Drive, but the Sheets API returned 403 PERMISSION_DENIED; its cells were not read. • Contents unknown. • Notion mining concept pages • Title search returned mining/KYC concept material. The pages are concepts or historical conversation records, not the exact approved proof package. • Concepts, not implementation or economic approval. • GitLab: LTReality-Social-Universe-Miracle-Network — https://gitlab.com/RainbowJobsMN/LTReality-Social-Universe-Miracle-Network • Read-only repository searches for Groth16, circom, verification_key, proof vector, storage proof, and compute proof returned no matches. • No matching repository files found in those searches. • OneDrive • Searches for the same proof terms and reward economics returned no files. • No matching files in those searches. • Linear • Read-only search for proof, mining, reward, and storage issues returned no matching issues. • No matching issues found. • X and xAI • X returned 401 Unauthorized; xAI returned 400 with an incorrect-key error. Reauthorization context reported no reconnect option for either. No X results or Grok model response were obtained. • Unavailable; no AI-generated findings. • GitHub and GitLab source-control • These attached routes remained connect rather than ready, so repository searches were not available through them. The ready GitLab read-only API route was used instead. • Not callable in this review. • Cursor • The connected integration exposes coding-agent/team operations, not a suitable evidence-search source. No agent was launched and no usage data was read. • Not used; avoids unnecessary agent action and possible cost. • Drive search results were relevance-based and some queries were paginated or rate-limited. The exact .circom filename search returned no files; queries containing .r1cs, .wasm, and .zkey surfaced email/metadata records rather than confirmed proof artifacts. This is not a complete scan of every file in every connected account.

Fair and sustainable commercial direction

The source review does not substantiate the earnings ranges or return claims in the research materials. The earlier report review classifies the cited DePIN, LP, staking, trading, flash-loan, and token-appreciation estimates as unsupported or requiring independent financial/security review. Those figures should not be repeated as expected earnings.

The most defensible proposal to validate, not an approved business plan, is:

1. Start with customer-funded utility. Evaluate a narrowly scoped storage or compute service only where customers demonstrably pay for the service. A

token emission, NFT mint, internal price, or deposit is not service revenue.
2. Calculate real contribution margin. Measure collected customer receipts minus storage/compute, bandwidth/egress, payment, support, fraud, security, compliance, and other directly attributable costs. Do not subsidize rewards from an assumed future token price.
3. Fund any later contributor pool from realized, attributable surplus. Publish eligibility, measurement, disputes, caps, and accounting before activation. Separate user rewards from reserves, liabilities, and treasury balances. No rates or allocations are selected here.
4. Pay only for independently verifiable accepted work. The proof must bind the trusted assignment and exact work statement. A storage challenge proves only its approved statement at that time; it does not prove perpetual retention, replication, data rights, or reserve value.
5. Protect participants. Use clear opt-in, privacy-minimizing evidence, disclosed costs and risks, accessible disputes, and no guaranteed earnings language. Do not require users to buy hardware or stake funds merely to make unverified rewards appear profitable.

Before considering any pilot, compare candidates using the same measured scorecard: paying-customer demand, collected revenue, variable cost per accepted unit, proof/verification cost, uptime and service quality, privacy and rights burden, fraud/dispute rate, and net margin after all attributable costs. Run bounded simulations first; no transaction or provider setup was performed for this task.

Exact Task 81 unblockers

1. Owner-defined compute workload, input commitment, deterministic output, exact unit derivation, and stable work identity.
2. Owner-defined storage dataset/root construction, hash/tree and chunk rules, fresh challenge policy, index/path encoding, and exact claim being proven.
3. Explicit approval by exact version and hash of the verifier, circuit, public verifying key, statement, and byte-level public-input encoding.
4. Independently sourced positive compute and storage vectors with complete bytes, source revision, provenance, hashes, and expected outcomes.
5. Reproducible tests showing genuine positives pass; altered inputs, chunks, indices, roots, paths, and proof bytes fail; and a second valid encoding of consumed work is rejected.
6. Existing economic sentinels remain unchanged, and genuine-proof positives have no minting, redeemable-balance, or backing side effects.

Until those items exist and are approved, the correct Task 81 status is BLOCKED. A successful proof would verify only the approved statement; it would not establish revenue, backing, customer demand, or guaranteed earnings.

Source boundaries

- Task 81 policy — report-template-proof-validation.md
- Candidate proof-system design — candidate-proof-system-design.md
- Comprehensive crypto report review — comprehensive-crypto-report-review.md
- Cloud-source search log — proof-artifact-cloud-search.md
- AI-agent workflow and completion guide — report-template-proof-agent-guide.md